Raquel Forés Toribio

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Employment

2024 – present **Postdoctoral Researcher** CCAPP Fellow
Department of Astronomy, The Ohio State University, OH, USA.

2020 – 2024 **Graduate grant-assisted student** Supervisor: Jose Antonio Muñoz Lozano
Departament d'Astronomia i Astrofísica, Universitat de València, València, Spain.

Education

2020 – 2024 **PhD in Physics,** Cum Laude, International mention
Universitat de València, València, Spain.

2018 – 2020 **Master's Degree in Astrophysics,** Average mark: 9.84/10
Universidad de La Laguna, Tenerife, Spain.

2014 – 2018 **Bachelor Degree in Physics,** Average mark: 9.35/10
Universitat de València, València, Spain.

Scientific Products

Journal Articles

1. B. JoHantgen, D. M. Rowan, **R. Forés-Toribio**, *et al.*, "A Systematic Search for Big Dippers in ASAS-SN," *OJAp*, vol. 9, Feb. 2026. DOI: 10.33232/001c.156224.
2. E. Mediavilla, J. Jiménez-Vicente, C. Fian, *et al.*, "Evidence for Substructure on the Inner Broad Line Region of the Lensed Quasar SDSS J1004+4112 from the Recurrence of Microlensing in the C IV Emission Line," *ApJ*, vol. 997, no. 2, 274, p. 274, Feb. 2026. DOI: 10.3847/1538-4357/ae2c84.
3. S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. III. Thirty More Giant Planets," *ApJS*, vol. 280, no. 1, 30, Sep. 2025. DOI: 10.3847/1538-4365/aded0d.
4. **R. Forés-Toribio**, B. JoHantgen, C. S. Kochanek, *et al.*, "ASASSN-24fw: An 8-month long, 4.1 mag, optically achromatic and polarized dimming event," *OJAp*, vol. 8, 114, Aug. 2025. DOI: 10.33232/001c.143105.
5. Z. Essack, D. Dragomir, P. A. Dalba, *et al.*, "Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. VI. Confirmation of a Long-period Giant Planet Discovered with a Single TESS Transit," *AJ*, vol. 170, no. 1, 41, Jul. 2025. DOI: 10.3847/1538-3881/add88b.
6. T. R. Fairnington, J. Dong, C. X. Huang, *et al.*, "The eccentricity distribution of warm sub-Saturns in TESS," *MNRAS*, vol. 540, no. 1, Jun. 2025. DOI: 10.1093/mnras/staf759.
7. J. Ehrhardt, L. Thomas, H. Kellermann, *et al.*, "Confirmation of four hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES," *A&A*, vol. 692, A220, Dec. 2024. DOI: 10.1051/0004-6361/202451404.
8. **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, "Microlensing analysis of 14.5-year light curves in SDSS J1004+4112: Quasar accretion disk size and intracluster stellar mass fraction," *A&A*, vol. 691, A97, Nov. 2024. DOI: 10.1051/0004-6361/202347378.
9. C. Fian, J. Jiménez-Vicente, E. Mediavilla, *et al.*, "First Direct Evidence for Keplerian Rotation in Quasar Inner Broad-line Regions," *ApJL*, vol. 972, L7, Sep. 2024. DOI: 10.3847/2041-8213/ad654d.

10 C. Fian, J. A. Muñoz, J. Jiménez-Vicente, *et al.*, "Revealing the inner workings of the lensed quasar SDSS J1339+1310: Insights from microlensing analysis," *A&A*, vol. 689, A129, Sep. 2024. DOI: 10.1051/0004-6361/202450151.

11 **R. Forés-Toribio**, E. Mediavilla, J. A. Muñoz, J. Jiménez-Vicente, C. Fian, and C. del Burgo, "Single-epoch and Differential Astrometric Microlensing of Quasars," *ApJ*, vol. 970, 11, Jul. 2024. DOI: 10.3847/1538-4357/ad49a6.

12 B. J. Hord, E. M. R. Kempton, T. M. Evans-Soma, *et al.*, "Identification of the Top TESS Objects of Interest for Atmospheric Characterization of Transiting Exoplanets with JWST," *AJ*, vol. 167, 233, May 2024. DOI: 10.3847/1538-3881/ad3068.

13 C. Fian, J. A. Muñoz, **R. Forés-Toribio**, *et al.*, "Probing the structure of the lensed quasar SDSS J1004+4112 through microlensing analysis of spectroscopic data," *A&A*, vol. 682, A57, Feb. 2024. DOI: 10.1051/0004-6361/202347382.

14 F. Rescigno, G. Hébrard, A. Vanderburg, *et al.*, "A hot mini-Neptune and a temperate, highly eccentric sub-Saturn around the bright K-dwarf TOI-2134," *MNRAS*, vol. 527, pp. 5385–5407, Jan. 2024. DOI: 10.1093/mnras/stad3255.

15 S. Yoshida, S. Vissapragada, D. W. Latham, *et al.*, "TESS Spots a Super-puff: The Remarkably Low Density of TOI-1420b," *AJ*, vol. 166, 181, Nov. 2023. DOI: 10.3847/1538-3881/acf858.

16 S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. II. Twenty New Giant Planets," *ApJS*, vol. 265, 1, Mar. 2023. DOI: 10.3847/1538-4365/aca286.

17 **R. Forés-Toribio**, J. A. Muñoz, C. S. Kochanek, and E. Mediavilla, "A Mass Model for the Lensing Cluster SDSS J1004+4112: Constraints from the Third Time Delay," *ApJ*, vol. 937, 35, Sep. 2022. DOI: 10.3847/1538-4357/ac8c40.

18 J. A. Muñoz, C. S. Kochanek, J. Fohlmeister, J. Wambsganss, E. Falco, and **R. Forés-Toribio**, "The Longest Delay: A 14.5 yr Campaign to Determine the Third Time Delay in the Lensing Cluster SDSS J1004+4112," *ApJ*, vol. 937, 34, Sep. 2022. DOI: 10.3847/1538-4357/ac8877.

19 S. W. Yee, J. N. Winn, J. D. Hartman, *et al.*, "The TESS Grand Unified Hot Jupiter Survey. I. Ten TESS Planets," *AJ*, vol. 164, 70, Aug. 2022. DOI: 10.3847/1538-3881/ac73ff.

20 J. Dong, C. X. Huang, G. Zhou, *et al.*, "NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star," *ApJL*, vol. 926, L7, Feb. 2022. DOI: 10.3847/2041-8213/ac4da0.

Conferences

1 **R. Forés-Toribio**, B. JoHantgen, C. S. Kochanek, *et al.*, *ASASSN-24fw in the context of long and deep stellar dimmings in time-domain surveys (contributed talk), The 16th Gaia Science Alerts and ACME Time-Domain Workshop, Tenerife, Spain*, Sep. 2025.

2 **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Quasar microlensing in galaxy clusters as a complementary approach to intracluster light (contributed talk), XVI Reunión Científica de la SEA, Granada, Spain*, Jul. 2024.

3 **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Constraining the stellar mass fraction and the quasar accretion disk size of SDSS J1004+4112 with microlensing (poster), 26th International Microlensing Conference, Livermore, California, USA (online)*, Jan. 2024.

4 **R. Forés-Toribio**, J. A. Muñoz, C. Fian, J. Jiménez-Vicente, and E. Mediavilla, *Stellar mass fraction and quasar accretion disk size in SDSS J1004+4112 from photometric follow-up (contributed talk), IAU Symposium 381: Strong Gravitational Lensing in the Era of Big Data, Otranto, Italy*, Jun. 2023.

5. **R. Forés-Toribio**, *Single epoch astrometric microlensing in lensed quasars (contributed talk), 25th International Microlensing Conference, Paris, France*, Sep. 2022.

6. **R. Forés-Toribio**, J. A. Muñoz, C. S. Kochanek, and E. Mediavilla, *Remodelling the lensing cluster SDSS J1004+4112 (poster), European Astronomical Society Annual Meeting, Valencia, Spain*, Jun. 2022.

Invited Talks, Colloquiums and Seminars

1. **R. Forés-Toribio**, *ASASSN-24fw in the context of long and deep stellar dimmings in time-domain surveys, CCAPP Symposium 2025, Center for Cosmology and AstroParticle Physics, The Ohio State University, Columbus, Ohio, USA*, Sep. 2025.

2. **R. Forés-Toribio**, *ASASSN-24fw: a long, deep, achromatic and polarized stellar eclipse, Ball State University Colloquium, Muncie, Indiana, USA*, Sep. 2025.

3. **R. Forés-Toribio**, *Lentes gravitatorias y su utilidad para la astronomía: El sistema B0445+123 y las observaciones con GTC, Invited talk to the GTC staff, Tenerife, Spain*, Jan. 2025.

Achievements

Grants

06/2021 – 09/2024 — **Graduate grant "Atracció de Talent"**, Universitat de València.

09/2020 – 06/2021 — **Graduate grant "ACIF"**, Generalitat Valenciana.

11/2017 – 07/2018 — **Collaboration grant** (Undergraduate research), Ministerio de Educación y Ciencia.

Prizes

2025 — **Premio Extraordinario de Doctorado del curso 2024/2025** *(Thesis Extraordinary Prize academic year 2024/2025).*
Awarded by Universitat de València, Valencia, Spain.

2024 — **Premi Mavi Dolç i Gastaldo 2024 a l'ús del valencià i la qualitat lingüística en les tesis doctorals** *(Mavi Dolç and Gastaldo Prize 2024 for the use of Valencian and linguistic quality in doctoral theses).*
Awarded by Universitat de València, Valencia, Spain.

2020 — **Premio Extraordinario en la Titulación de Máster Universitario en Astrofísica** *(Extraordinary Prize for the Master's Degree in Astrophysics).*
Awarded by Universidad de La Laguna, Tenerife, Spain.

Premio de Excelencia de Máster en la rama de conocimiento de Ciencias *(Prize of Excellence for the Master in the branch of knowledge in Sciences).*
Awarded by Universidad de La Laguna, Tenerife, Spain.

Competitive research projects

01/09/2025 – present — **PID2024-160091NB-C32 of the Ministerio de Ciencia e Innovación**. Member of the working group.

01/09/2021 – 31/08/2025 — **PID2020-118687GB-C32 of the Ministerio de Ciencia e Innovación**. Member of the working group.

01/10/2020 – 29/06/2021 — **AYA2016-79014-C3-3-P of the Ministerio de Ciencia e Innovación**. Member of the working group.

Skills

Languages — Strong reading, writing and speaking competencies for English (Advanced Cambridge English Level), Spanish (native) and Catalan (native).

Coding — Good knowledge of Python, FORTRAN and LaTeX.
Some knowledge of Mathematica, Julia, C++ and IRAF.

Miscellaneous

International stays

08/2022 – 11/2022 — **Department of Astronomy at the Ohio State University**
with Prof. Christopher Kochanek.

Teaching experience

Course 2024-25 — **Observational Techniques in Astronomy and Photometric Observations, 15 hours**. Experto Universitario en Astrofísica Observacional (Postgraduate course), Universitat de València.

Course 2023-24 — **Observational Techniques in Astronomy and Photometric Observations, 15 hours**. Experto Universitario en Astrofísica Observacional (Postgraduate course), Universitat de València.

General Physics III (Electromagnetism and Optics), 15 hours. Degree in Physics, Universitat de València.

Course 2022-23 — **Mechanics Laboratory, 25 hours**. Degree in Physics, Universitat de València.

General Physics III (Electromagnetism and Optics), 30 hours. Degree in Physics, Universitat de València.

Course 2021-22 — **Mechanics Laboratory, 25 hours**. Degree in Physics, Universitat de València.

General Physics III (Electromagnetism and Optics), 30 hours. Degree in Physics, Universitat de València.

Course 2020-21 — **Mechanics Laboratory, 50 hours**. Degree in Physics, Universitat de València.

Supervised works

Course 2023-24 — **Final Degree Project (co-supervised)**. Title: *Validación de candidatos a exoplanetas y determinación de parámetros orbitales (Exoplanet candidate validation and orbital parameters determination)* Author: Inés Pastor Espinosa. Degree in Physics, Universitat de València.

Course 2022-23 — **Final Degree Project (co-supervised)**. Title: *Seguimiento fotométrico de exoplanetas (Photometric follow-up of exoplanets)* Author: Paula Talavera Capilla. Degree in Physics, Universitat de València.

Referee experience

2024 – present — **Referee for the AAS journals**.

Observing proposals

2025 — **Request for observational time at Large Binocular Telescope**
as PI (observed).

2024 — **Request for observational time at Gran Telescopio Canarias**
as PI (observed).

Miscellaneous (continued)

- **Request for observational time at Gran Telescopio Canarias** as co-investigator (observed).

2023
- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

2022
- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

2021
- **Request for observational time at Gran Telescopio Canarias** as PI (awarded time).

Extracurricular activities

2024 – present
- **CCAPP seminar coordinator**. Weekly seminar series held by the Center for Cosmology and AstroParticle Physics (CCAPP).

2022 – present
- **Member of the Spanish Astronomical Society**.
- **Member of Astrobitos**. Blog of outreach in Spanish supported by AAS.

2020 – 2024
- **Member of the Department council**. Graduate students representative at the Departament d'Astronomia i Astrofísica, Universitat de València.

09/2023
- **Member of the organizing committee of E3F**. Escola Estiu Erasmus en Física (Student Summer School), Universitat de València.

Specialized training

2024
- **Responsible Conduct of Research**. Awarded by CITI program.

2023
- **Nocions essencials d'accessibilitat digital per a la docència** (Essential notions of digital accessibility for teaching). Awarded by Universitat de València.

2022
- **Julia for Scientists**. Awarded by Universitat de València.

2021
- **XXI Spanish Virtual Observatory School**. Awarded by Centro de Astrobiología.
- **The 5th Azarquiel School Of Astronomy**. Awarded by Azarquiel School.
- **Introduction to Machine Learning**. Awarded by Universitat de València.
- **Introduction to Statistics**. Awarded by Universitat de València.
- **Time-Domain Cosmology with Strong Gravitational Lensing**. Awarded by KAVLI IPMU.

2019
- **XII COEFIS** (Physics Students Congress). Awarded by Universidad de La Laguna.

2017
- **External practices at the Observatori Astronòmic de la Universitat de València**. Awarded by Universitat de València.